

Question 1 (20 Marks)

Development Plan for JU CSE Student Information System (SIS)

1. Project Deliverables

The project will create an SRS document. The project will create a system design document. The project will create a database design. The project will create source code. The project will create test reports. The project will create a user manual. The project will create the deployed Student Information System. These items make sure that every part of the project is written down and checked properly.

2. Project Interfaces

The Student Information System will interact with students, teachers, department staff and administrators through a web interface. It will also work with the university database and LDAP authentication for a way for users to log in and manage data. The Student Information System has these interfaces so students and teachers and department staff and administrators can talk to the Student Information System easily.

3. Development Methodology and Development Tools

The project follow the Scrum Agile methodology because of the requirements may change during the development and regular feedback is important. Development tools is include Visual Studio, ASP.NET Core, MS SQL Server, Git and Jira for project tracking. This combination supports faster development and continuous improvement.

4. Software Development Standards and Procedures

The team should follow the coding standard, code reviews, version control, documentation guidelines and software testing procedures. Following these standards to improves code quality, consistency and maintainability.

5. Development Process Map

The development process follow the sequence:

Requirement Analysis → System Design → Development → Testing → User Acceptance Testing (UAT) → Deployment → Maintenance.

This process ensure that the every phase is completed and verified before moving to the next stage.

6. Project Milestones

Major milestones are included:

- Requirements Approved
- System Design Completed
- Core Modules Developed
- System Testing Completed
- UAT Completed
- Final Deployment

These miletones help to monitor the project progress and keep the team on the schedule.

7. Team Organization and Responsibilities

- **Project Manager** – Plans and monitors the project.
- **Business Analyst** – Collects and manages requirements.
- **Developers** – Build the software.

- **QA Team** – Performs testing and quality assurance.
- **Database Administrator (DBA)** – Manages the database.
- **System Administrator** – Handles deployment and server maintenance.

Clearly defined roles improve coordination and accountability.

8. Required Development Facilities

The project requires the development computers, Visual Studio, MS SQL Server, Git repository, testing server, internet access and backup facilities. These resources support the efficient software development and testing.

9. Development Risks and Risk Management Activities

Possible risks include the changing requirements, developer resignation, schedule delays and security issues. The team will reduce these risks through the regular meeting to progress tracking, backup plans, code reviews, continuous testing and proper documentation.

10. Project Control Methods

The project will be controlled through weekly Scrum meeting, milestone reviews, progress reports, issue tracking and regular code reviews. These activities help identify problems early and keep the project on the track.

11. Project Cost Estimation

The budget will include developer salaries, software tools, server infrastructure, testing costs, training and maintenance expenses. Proper cost estimation helps manage resources efficiently and reduces the risk of budget overruns.

Conclusion

This Development Plan provides a clear roadmap for building the **JU CSE Student Information System**. By using the Scrum Agile, following software standards, managing risks and monitoring project progress, the team can deliver a reliable, secure and high-quality system on time.

Question 2 (20 Marks)

Software Quality Plan for JU CSE Student Information System (SIS)

1. Quality Goals

Quality Goal	Measurable Metric
Low defect rate	Critical defects = 0 before release
Good performance	Response time < 2 seconds for normal operations
High reliability	99% successful test cases
Good usability	At least 90% positive feedback in UAT
Secure system	No High/Critical security vulnerabilities

2. Planned Review Activities

Activity	Objective	Responsible Person	Timing	Deliverable
Requirements Review	Ensure requirements are complete and correct	Business Analyst & Project Manager	Before development	Approved SRS

Activity	Objective	Responsible Person	Timing	Deliverable
Design Review	Verify system architecture and design	System Architect	Before coding	Design Review Report
Code Review	Detect coding errors and improve code quality	Senior Developer	During development	Reviewed Source Code
Test Case Review	Ensure all requirements are covered	QA Lead	Before testing	Approved Test Cases

3. Software Testing Strategy

Testing Type	Objective	Responsible Person	Timing	Deliverable
Unit Testing	Test individual modules	Developer	During coding	Unit Test Report
Integration Testing	Verify module interaction	QA Team	After module integration	Integration Test Report
System Testing	Test the complete system	QA Team	Before UAT	System Test Report
Performance Testing	Check system speed and scalability	QA Team	Before release	Performance Test Report
Security Testing	Verify system security	Security Engineer	Before release	Security Test Report
User Acceptance Testing (UAT)	Confirm the system meets user needs	End Users & QA Team	Final stage	UAT Sign-off

4. Acceptance Testing Plan

- **Objective:** Ensure the system meets all business and user requirements.
- **Responsible Person:** End Users, Project Manager and QA Team.
- **Timing:** After successful System Testing.
- **Expected Deliverable:** UAT Report and User Acceptance Sign-off.

5. Configuration Management Plan

- **Objective:** Control software versions and project documents.
- **Responsible Person:** Configuration Manager.
- **Timing:** Throughout the project.
- **Expected Deliverable:** Version History, Change Log and Configuration Records.

6. Change Management Process

- **Objective:** Ensure that the all changes are reviewed, approved and documented before implementation.
- **Responsible Person:** Project Manager and the Change Control Board (CCB).
- **Timing:** Whenever a change requesting is submitted.
- **Expected Deliverable:** Approved Change Request and Updated Project Documents.

Conclusion

This Software Quality Plan helps to ensure that the **JU CSE SIS** is reliable, secure, and user-friendly. Regular reviews, comprehensive testing, configuration control and proper change management to improve the software quality and increase the chances of project success.

Question 3 (20 Marks)

Software Risk Analysis for JU CSE Student Information System (SIS)

Risk	Risk Category	Probability	Impact	Mitigation Strategy	Contingency Plan
1. Project schedule delay due to unexpected issues.	Scheduling	High	High	Prepare a realistic schedule and track progress regularly.	Revise the timeline and assign additional resources if needed.
2. Changing university requirements during development.	Requirements	High	High	Gather and validate requirements with stakeholders before development.	Update the project plan and prioritize the new requirements.
3. Senior developers leave the project.	Personnel	Medium	High	Maintain proper documentation and share knowledge among team members.	Hire replacements or redistribute tasks within the team.
4. LDAP/API integration problems.	Technology	Medium	High	Test the API early and maintain communication with the provider.	Use temporary authentication or integrate the API later.
5. Server or hardware procurement is delayed.	Resource	Medium	Medium	Order hardware early and monitor procurement progress.	Use cloud or temporary servers until the hardware arrives.
6. Database performance becomes slow during course registration.	Technology	Medium	High	Optimize database queries and perform load testing.	Upgrade server resources and tune database performance.
7. Security breach or unauthorized access to student data.	Security	Low	High	Implement strong authentication, encryption, and regular security testing.	Restore data from backups and apply security patches immediately.
8. Third-party vendor delays in delivering software or services.	Subcontracting	Low	Medium	Select reliable vendors and monitor delivery schedules.	Use an alternative vendor or an in-house solution if necessary.

Conclusion

By the identifying these risks early and preparing both **mitigation strategies** and **contingency plan**, the JU CSE SIS project team can reduce project delays, improve software quality and increase the chances of

successful the project completion.

Question 4 (15 Marks)

Risk Prioritization

Rank	Risk	Justification (Probability, Impact & Risk Exposure)
1	Two senior developers resign	Probability: Medium. Impact: Very High because development may slow down and knowledge is lost. Risk Exposure: Highest.
2	University changes grading policy	Probability: Medium. Impact: High since the grading module must be redesigned and retested. Risk Exposure: High.
3	LDAP authentication API is delayed	Probability: Medium. Impact: High because user login and authentication cannot be completed on time. Risk Exposure: Medium-High.
4	Server procurement is postponed	Probability: Low-Medium. Impact: Medium because deployment and performance testing may be delayed. Risk Exposure: Medium.
5	Mid-semester examination schedule changes	Probability: High. Impact: Low because it mainly affects project planning and testing dates, not the software itself. Risk Exposure: Lowest.

Conclusion

The resignation of senior developers is the highest priority because it can directly affect development process speed and software development quality. The examination schedule change is the lowest priority since it mainly requires the schedule adjustments and has the little impact on the overall system.

Question 5 (15 Marks)

Quality Improvement (As Quality Manager)

1. Which quality goals have failed?

The following quality goals has been failed:

- **Reliability** – 180 defects, including 35 critical defects, show that the software is not stable.
- **Performance** – A response time of more than **5 seconds** during course registration does not meet performance expectations.
- **Usability** – Poor feedback from User Acceptance Testing (UAT) indicates that the system is not user-friendly.

2. Which review activities should have detected these earlier?

These issues should has been identified through:

- **Requirements Review** – To ensure performance and usability requirements were clearly defined.
- **Code Review** – To detect coding errors before testing.
- **Test Case Review** – To verify that important test scenarios were covered.
- **Performance Testing** and **Usability Review** – To identify slow response times and user interface problems before UAT.

3. What corrective actions should be taken?

- Fix the all **35 critical defects** before release software.

- Optimize the database and application to reduce response time.
- Improve the user interface based on UAT feedback.
- Perform regression testing after every fix to ensure no new defects are introduced.

4. Which metrics should be monitored before release?

Before releasing the system monitor:

- **Defect Density** – Number of defects remaining in the software.
- **Response Time** – Ensure key operations complete within the target time.
- **Test Coverage** – Confirm all important features have been tested.
- **Defect Fix Rate** – Track how many reported defects have been resolved.

Conclusion

The software should only be released after the critical defects are fixed, performance is improved, usability issues are resolved and all quality metrics meet the required standards.

Question 6 (10 Marks)

Software Risk Management Process for JU CSE SIS

The JU CSE SIS project team should manage risks throughout the project by following these steps:

1. Risk Identification

The team should identify the possible risks at the beginning of the project. For the JU CSE SIS, examples included changing requirements, shortage of skilled developers, data security issues and delays in project completion.

2. Risk Assessment

After identifying the risks the team should evaluate their **likelihood** and the **impact**. For example a data security breach has a high impact because of it could expose student personal and academic information.

3. Risk Planning

The team should prepare a plan for handling each major risk. For example regular data backup, secure authentication and realistic project schedule can reduce the effect of potential risks.

4. Risk Mitigation

The planned action should be implemented to reduce the chance of the risk occurring. For example, frequent code reviews, continuous testing and developer training can minimize the software defects and to improve the quality.

5. Risk Monitoring

Risk should be monitored throughout the project development phase. The team should follow the regularly review project progress, budget, schedule and newly identified risks to ensure that they remain under control.

6. Risk Review

At the end of each project phase the team should review how risks were managed and update the risk register if it is necessary. For example after completing student registration module, the team

can evaluate whether the planned security measures were effective and identify any new risks for the next phase.

Conclusion

By following these six steps, the JU CSE SIS team can identify risks early, reduce their impact, and improve the chances of delivering a secure, reliable and successful Student Information System.

Conceptual Questions (Understanding Test)

1. Why is a Development Plan insufficient without a Quality Plan?

Answer:

A Development Plan focuses on the how software will be developed and delivered. However it does not guarantee to the software will work correctly. A Quality Plan ensures proper testing, reviews and quality check so the final system is reliable and meets user expectations.

2. Explain the difference between project milestones and project deliverables.

Answer:

A **milestone** is an important the checkpoint that shows progress in the project, such as completing the design phase. A **deliverable** is the actual output of the project, such as a requirements document, software module or final application.

3. Why should quality goals be measurable rather than qualitative?

Answer:

Software Measurable quality goals make it easy to check whether the software meets the required standard. For example saying the “response time should be less than 2 seconds” is much clear than simply saying “the system should be fast.”

4. Explain the difference between risk mitigation and contingency planning.

Answer:

Risk mitigation means taking the steps in advance to reduce the chance of a risk occurring. **Contingency planning** is preparing a backup plan so the team knows what to do if the risk actually happens.

5. Why is configuration management considered a part of a Software Quality Plan?

Answer:

Configuration management keeps track of different versions of the software and project documents. It helps the team avoid conflicts, recover previous versions if needed, and maintain consistent software quality throughout development.

6. Which development plan components are most affected when project requirements change?

Answer:

The project **schedule**, **budget**, and **resource allocation** are affected the most. New or changed requirements usually require extra time, additional work and sometimes more team members.

7. Explain how poor requirement management can increase project risk.

Answer: If requirements are not managed properly, the project may include unnecessary features or miss important ones. This can increase costs, delay delivery, and reduce customer satisfaction.

8. Why should acceptance testing criteria be defined before software development begins?

Answer: Defining acceptance criteria early gives everyone a clear understanding of what the final software should achieve. It also prevents confusion and disagreements when the project is completed.

9. Explain the relationship between project control methods and software quality.

Answer: Project control methods, such as regular progress reviews, code inspections, and testing, help identify problems early. Fixing issues during development improves the overall quality of the software.

10. Which software risks are most critical for a university-wide Student Information System and why?

Answer: The most critical risks are **data security** and **system scalability**. The system stores sensitive student information, so strong security is essential. It must also handle thousands of users at the same time, especially during course registration and result publication, without slowing down or crashing.

Higher-Order (Critical Thinking) Questions

Suppose JU decides to expand the SIS for all faculties after six months of development. How should the Development Plan and Quality Plan be revised?

Answer:

If JU decides to expand the SIS for **all faculties** after six months, both the **Development Plan** and **Quality Plan** should be updated.

- **Development Plan:** Revise the project scope, timeline, resource allocation, and budget to support the additional faculties and new requirements.
- **Quality Plan:** Add more testing activities, increase performance and security testing, and conduct user acceptance testing (UAT) with representatives from different faculties.

Conclusion: Updating both plans ensures the expanded system remains reliable, meets the needs of all faculties and maintains high software quality.

Compare Agile Scrum and Waterfall for developing JU CSE SIS. Which would you recommend? Justify your answer considering quality assurance and risk management.

Answer:

Agile Scrum	Waterfall
Development is done in small iterations (sprints).	Development follows a fixed sequence of phases.
Requirements can be changed during the project.	Changes are difficult once development starts.
Testing is done continuously in every sprint.	Testing is mostly done after development is completed.
Risks are identified and handled early.	Risks may be discovered later in the project.

Recommendation:

I would recommend **Agile Scrum** for the JU CSE SIS project. Since university requirements may change during development, Scrum provides flexibility to adapt quickly. Continuous testing improves **quality assurance**, while regular reviews and feedback help identify and reduce **risks** early. This increases the chances of delivering a high-quality system on time.

If the project budget is reduced by 25% while the deadline remains unchanged which Development Plan elements should be revised first? Explain your reasoning.

Answer:

If the project budget is reduced by **25%** but the deadline stays the same, these **Development Plan** elements should be revised first:

1. **Resource Allocation:** Reassign team members and reduce non-essential resources to stay within the new budget.
2. **Project Scope:** Delay or remove low-priority features so the team can focus on the most important functions.
3. **Risk Management Plan** – Update the risk plan because a lower budget increases the chances of delays, quality issues, and resource shortages.

Reasoning: Revising these elements first helps control costs while keeping the project on schedule and ensuring that the most important objectives are still achieved.

Propose three quality metrics that should be monitored during development and explain how they support project success.

Answer:

To ensure the success of the JU CSE SIS project, the following three quality metrics should be monitored during development:

1. **Defect Density:** Measures the number of bugs found in the software. A lower defect density means better software quality and fewer issues after release.
2. **Test Coverage:** Shows how much of the system has been tested. Higher test coverage helps detect problems early and improves software reliability.
3. **Requirements Coverage:** Checks whether all project requirements have been implemented and tested. This ensures the system meets user needs and reduces the chance of missing important features.

Conclusion: Monitoring these metrics helps improve software quality, reduce risks, and ensure the project is delivered successfully.

Based on Boehm and Ross's Top 10 Software Risk Items identify the three significant risks for JU CSE SIS and justify why they pose the greatest threat to project success.

Answer:

According to Boehm and Ross's Top 10 Software Risk Items, the three most significant risks for the JU CSE SIS project are:

1. **Personnel Shortfalls:** A lack of skilled developers or testers can delay the project and reduce software quality.
2. **Unrealistic Schedules and Budgets:** Tight deadlines and limited budgets may lead to incomplete features, poor testing, and higher project failure risk.
3. **Developing the Wrong User Interface:** If the system is difficult to use, students, teachers and staff may not accept or use it effectively.

Conclusion: These risks are the most critical because they directly affect the project's quality, timeline, and user satisfaction.